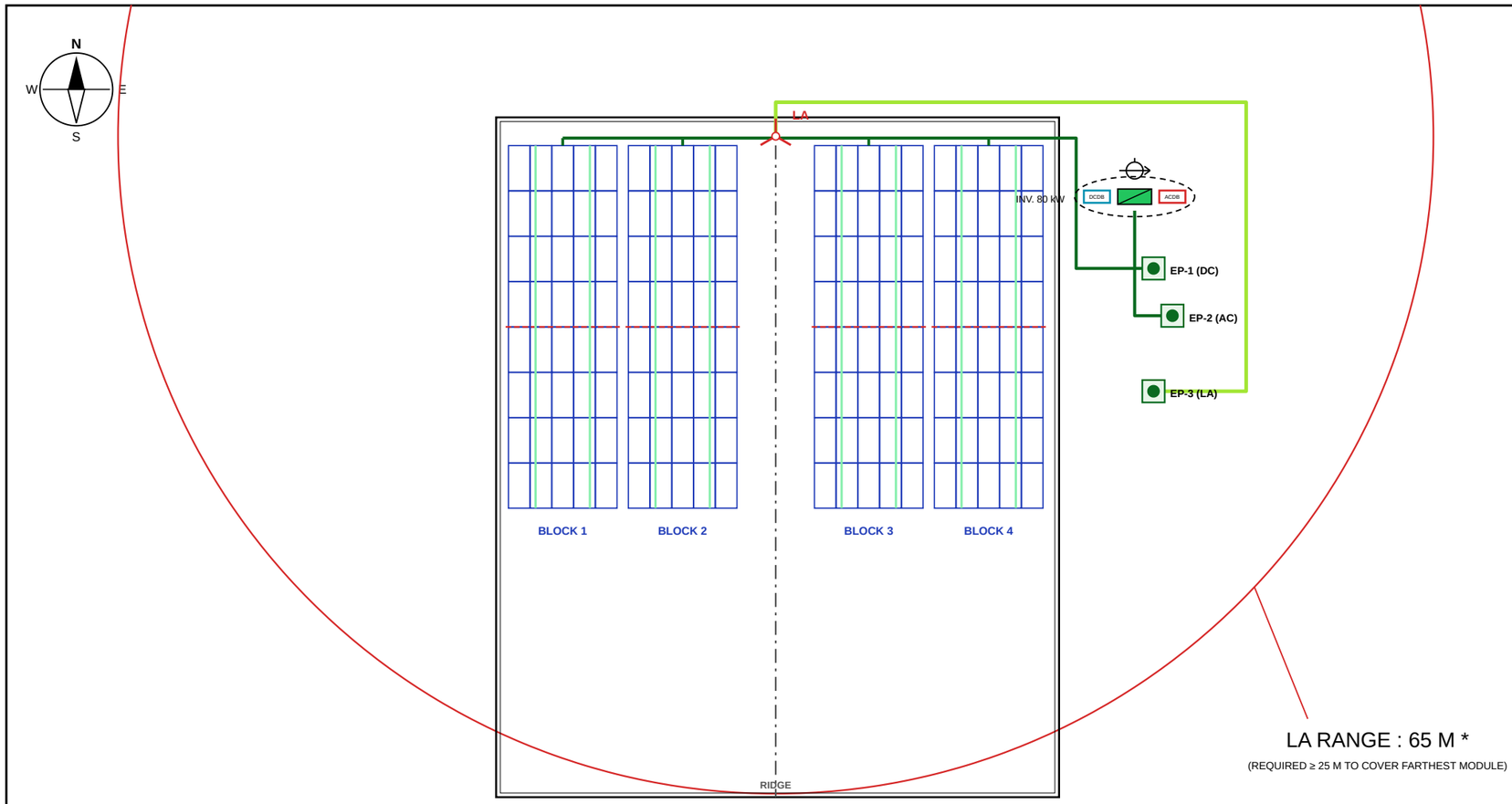


LEGENDS

	SOLAR MODULES
	STRING INVERTER
ACDB	AC DISTRIBUTION BOX
DCDB	DC DISTRIBUTION BOX
	EARTH CABLE (FOR DC & AC)
	EARTH CABLE (FOR LA)
	EARTHING PIT
	LIGHTNING ARRESTER
	MODULE TO MODULE EARTHING CABLE
	MMS TO MMS EARTHING CABLE
	FIRE-EXTINGUISHER

SYSTEM DESCRIPTION

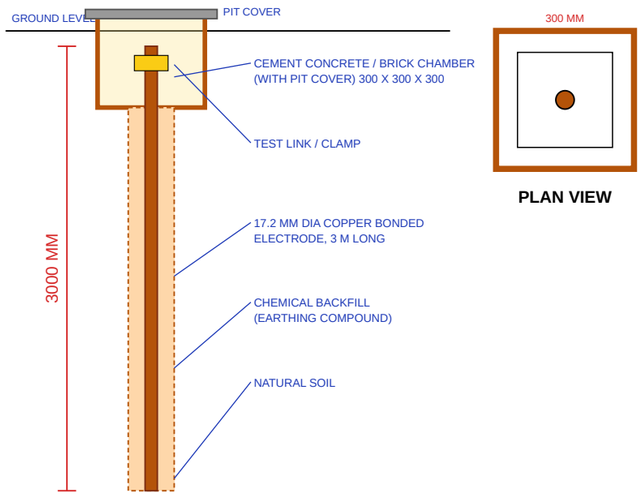
SITE LOCATION	— (LAT, LONG TO BE ADDED)
DC CAPACITY	100.00 kWp
TOTAL PLANT CAPACITY (AC)	80 kW
DC / AC RATIO	1.25
MODULE CAPACITY	625 Wp (ADANI N-TOPCon)
MODULE DIMENSION (L X W X H)	2382 X 1134 X 30 mm *
TOTAL PV MODULES	160 Nos.
INVERTER CAPACITY	80 kW (GOODWE GW80K-MT) 1 No.
AZIMUTH / TILT	EAST-WEST (90° / 270°), ROOF SLOPE

GENERAL NOTES

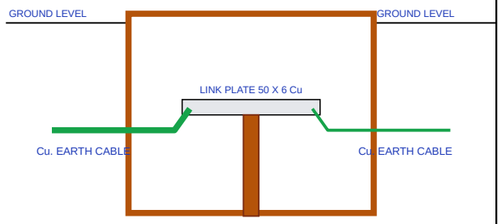
1. PV YARD EARTHING GRID WITH 16 SQ MM Cu EARTHING CABLE.
2. INVERTER & ACDB EARTHING WITH 25 SQ MM Cu EARTHING CABLE.
3. LA EARTHING WITH 50 SQ MM Cu CABLE, KEPT ≥ 1 m FROM DC / AC CABLES.
4. TOTAL EARTH RESISTANCE OF EARTHING SYSTEM SHOULD BE LESS THAN 1 Ω.
5. TOTAL 3 NOS. EARTH ELECTRODES: EP-1 DC, EP-2 AC, EP-3 LA.
6. MIN. DISTANCE BETWEEN EARTH PITS 3 m; 6 m PREFERRED.
7. 17.2 MM DIA x 3 m COPPER BONDED ELECTRODE WITH CHEMICAL BACKFILL.
8. MODULE TO MODULE EARTHING WITH 1C x 6 SQ MM Cu CABLE.
9. EARTH PIT LOCATION IS INDICATIVE; MAY VARY PER SITE CONDITIONS.
10. AFTER JOINT, CABLE SLEEVE SHOULD BE APPLIED.
11. ALL EQUIPMENT SHALL BE CONNECTED TO THE NEAREST EARTH PIT, WHETHER SHOWN IN THE DRAWING OR NOT.
12. PIT INTERCONNECTION AS PER CEIG APPROVAL.

* VALUES MARKED * TO BE CONFIRMED ON SITE / LA DATASHEET.

EARTHING AS PER IS 3043:2018



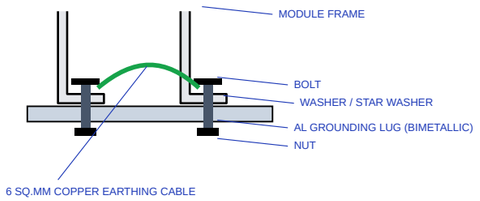
SECTION VIEW OF EARTH PIT



EQUIPMENT EARTHING CONNECTION
DETAIL FOR DC, AC & LA EARTH PITS

EARTH PIT SCHEDULE
EP-1 (DC) : PV STRUCTURE / MODULE FRAMES, DCDB, 16 SQ MM Cu
EP-2 (AC) : INVERTER PE & BODY, ACDB, AC SPD, CABLE ARMOUR, 25 SQ MM Cu
EP-3 (LA) : ESE LIGHTNING ARRESTER, 50 SQ MM Cu, INSULATED SPACERS, TEST LINK 1 m ABOVE GL

PIT SPACING (FROM LAYOUT IMAGE) ≈ 2.5 – 4 m *
PREFERRED ≥ 6 m (2 x ELECTRODE LENGTH) — MOVE EP-3 AWAY FROM EP-1 / EP-2 WHERE SITE ALLOWS.



MODULE GROUNDING DETAIL

CUSTOMER SIGN AND STAMP:

REV.	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APPROVED BY
R0	27-10-26	FIRST ISSUE			

CLIENT:	**JAIN GRANITE & MARBLE**
EPC CONTRACTOR:	**NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara
DESIGN BY:	**NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara

DRAWING NO.: 03

DRAWING NAME: **EARTHING LAYOUT**

PROJECT TITLE: 100.00 kWp (DC)_ROOFTOP SOLAR PLANT

RELEASED FOR: ☐ PRELIMINARY ☐ TENDER ☐ REVISION ☐ FABRICATION ☐ CONSTRUCTION ■ APPROVAL

SCALE: **NTS**

DESIGN BY	DRAWN BY	CHECKED BY	APPROVED BY	PAPER SIZE	SHEET NO.
27-10-26	27-10-26	27-10-26	27-10-26	**A3**	**01 OF 01**